Swiggy Food Delivery Data Engineering Project

Using Microsoft Fabric

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1. Objective

The objective of this project is to design and implement an end-to-end data engineering solution using Microsoft Fabric, starting from raw data ingestion to business insights generation.

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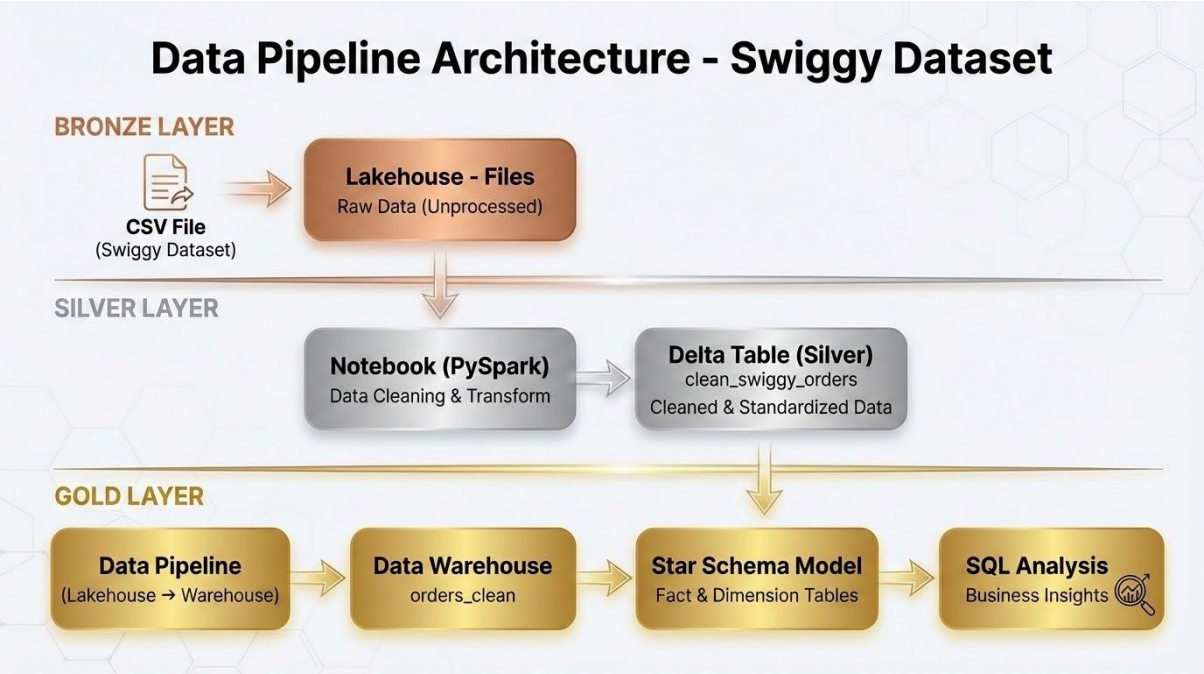
2. Architecture Overview

The project follows a **Medallion Architecture**:

Bronze → Raw Data
Silver → Cleaned Data
Gold → Business-ready Data

Data Flow:
CSV → Lakehouse → Notebook → Delta Table → Pipeline → Data Warehouse → Star Schema → SQL Analysis



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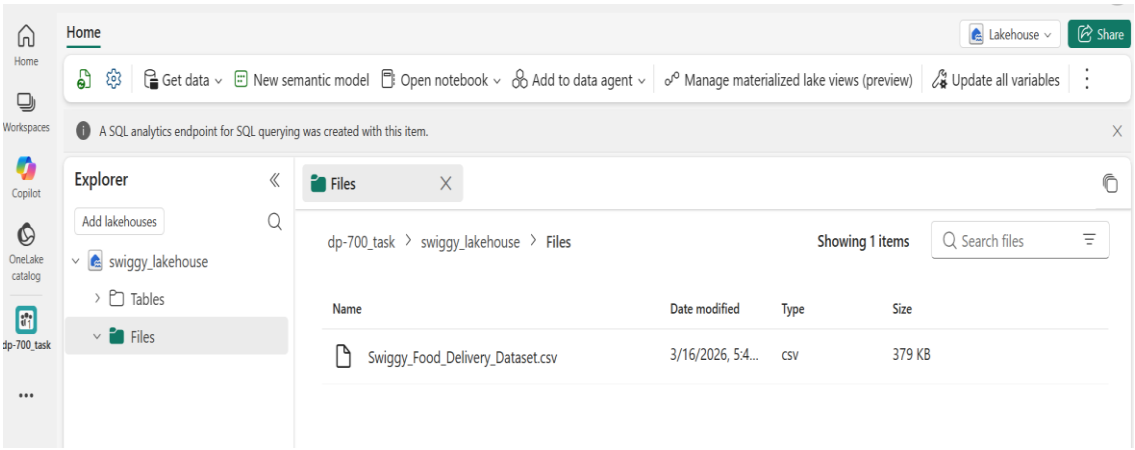
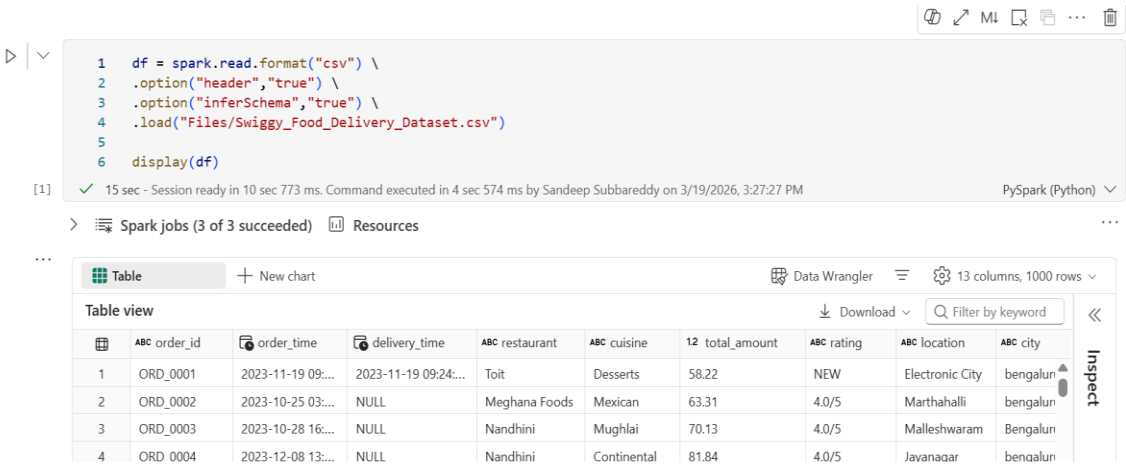
3. Data Ingestion

- Loaded Swiggy dataset (CSV) into Fabric Lakehouse (Files section)



- Accessed data using PySpark notebook



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4. Data Cleaning & Transformation

Performed data cleaning using PySpark as per requirements

Handling Missing Values

- Replaced null values in categorical columns with "Unknown"
- Handled missing numeric values using default values

String Standardization

- Applied trim(), initcap(), regexp_replace()
- Corrected inconsistent city names

Rating Transformation

- Converted rating from string format (e.g., "4.5/5") to numeric

- Created rating_numeric column

Data Validation

- Ensured total_amount contains only positive values

Conditional Logic

- Set delivery_time and distance_km to NULL when online_order = 'No'

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5. Delta Table Creation

- Stored cleaned data in Lakehouse as Delta table: **clean_swiggy_orders**

```
df.write.format("delta") \
  .mode("overwrite") \
  .option("overwriteSchema", "true") \
  .saveAsTable("Cleaned_swiggy_orders")

df.show()
```

[20] ✓ 5 sec - Command executed in 5 sec 734 ms by Sandeep Subbareddy on 3/19/2026, 3:31:32 PM PySpark (Python)

Spark jobs (11 of 11 succeeded) Resources

```
+--------+-------------------+-------------------+----------------+------------+------------+-------+---------------+---------+----
-------+----------+------------+-----------+-------------+
|order_id|         order_time|      delivery_time|      restaurant|     cuisine|total_amount| rating|       location|
city|ordered_qty|book_table|online_order|distance_km|rating_numeric|
+--------+-------------------+-------------------+----------------+------------+------------+-------+---------------+---------+----
```

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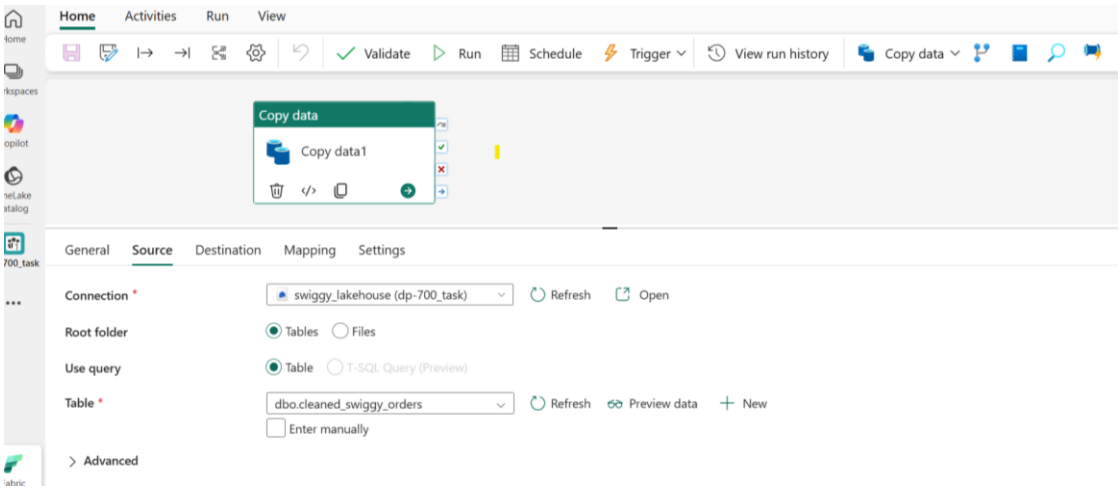
6. Data Pipeline

- Created a data pipeline to move cleaned data from Lakehouse to Data Warehouse
- Used Copy Activity for data transfer



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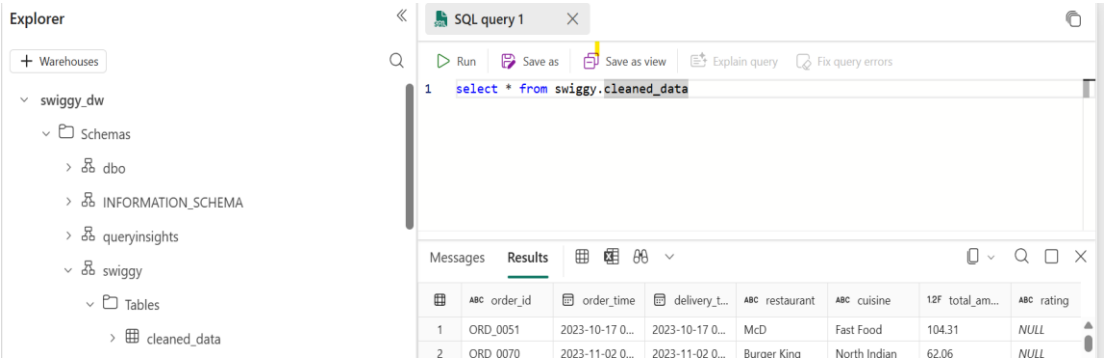
7. Data Warehouse

- Loaded cleaned data into Fabric Data Warehouse
- Created structured table:
  cleaned_data



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8. Data Modeling (Star Schema)

Fact Table

Fact_Orders

- order_id
- restaurant_key
- location_key
- time_key
- total_amount
- ordered_qty
- distance_km
- rating_numeric

Dimension Tables

Dim_Restaurant

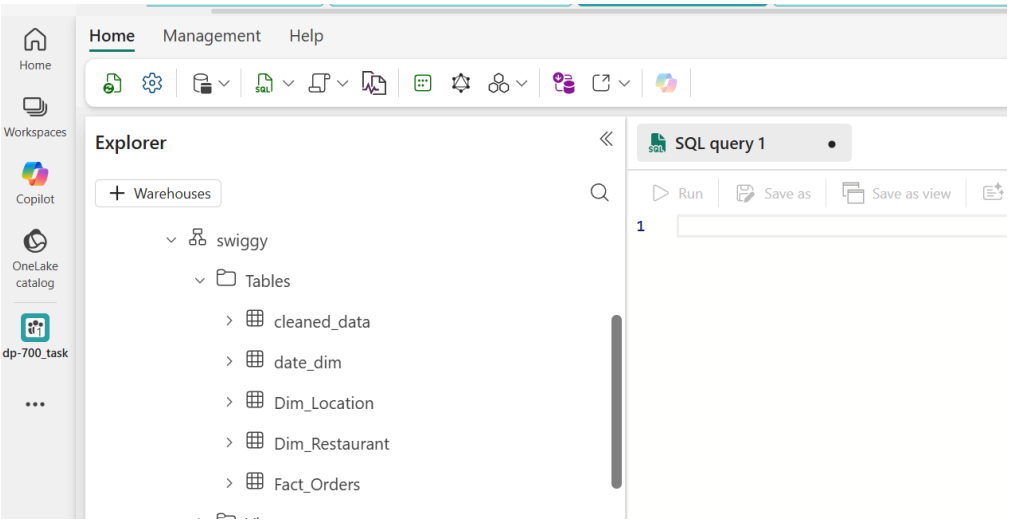
- restaurant_key
- restaurant_name
- cuisine_type

Dim_Location

- location_key
- area_name
- city_name

Dim_Time

- time_key
- full_timestamp
- hour
- day_name
- is_weekend

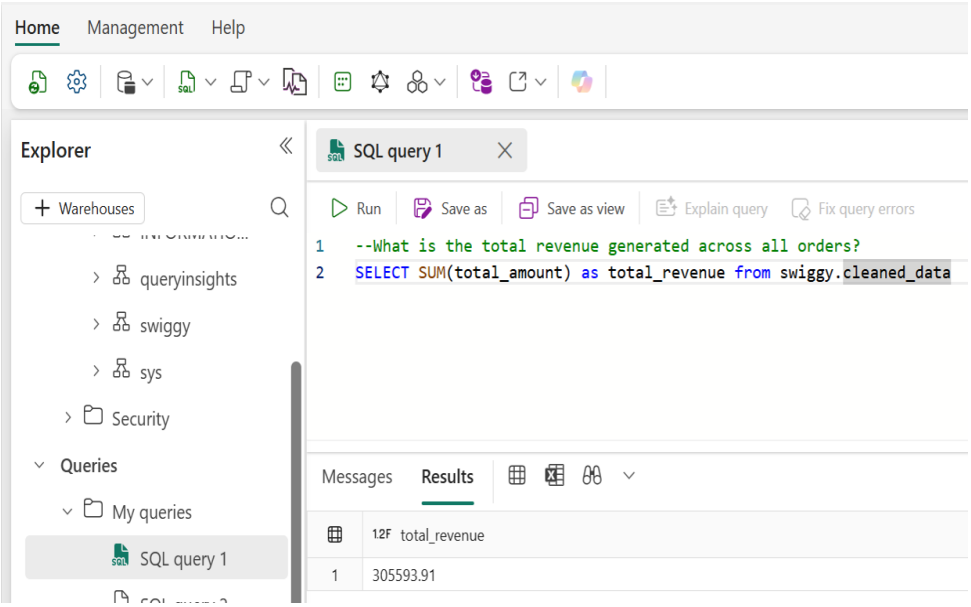


9. Data Analysis

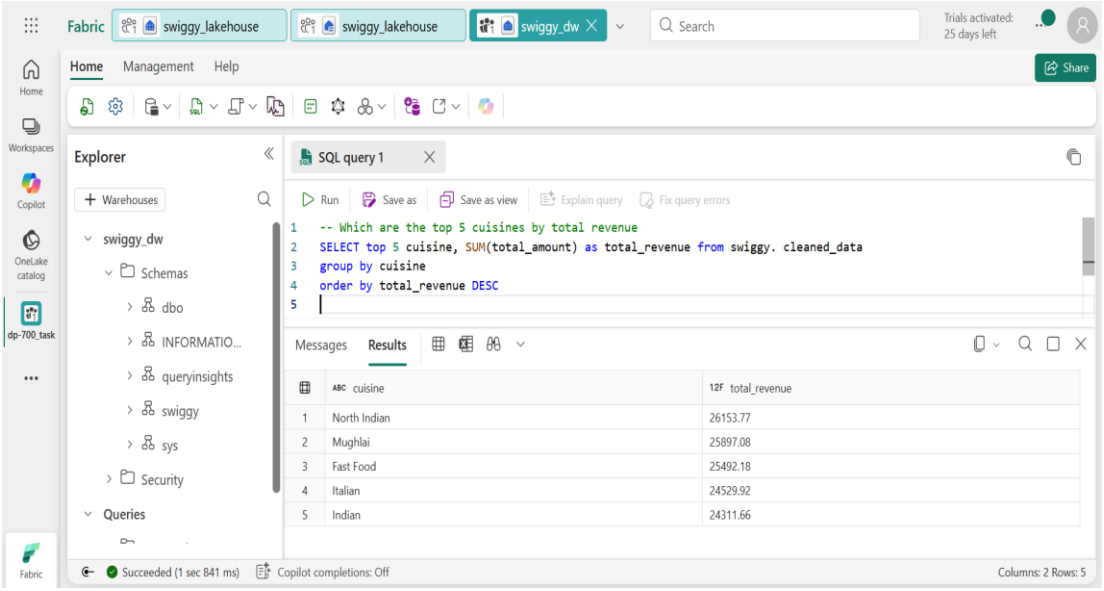
Performed SQL-based analysis as per business requirements

Category: Sales & Revenue

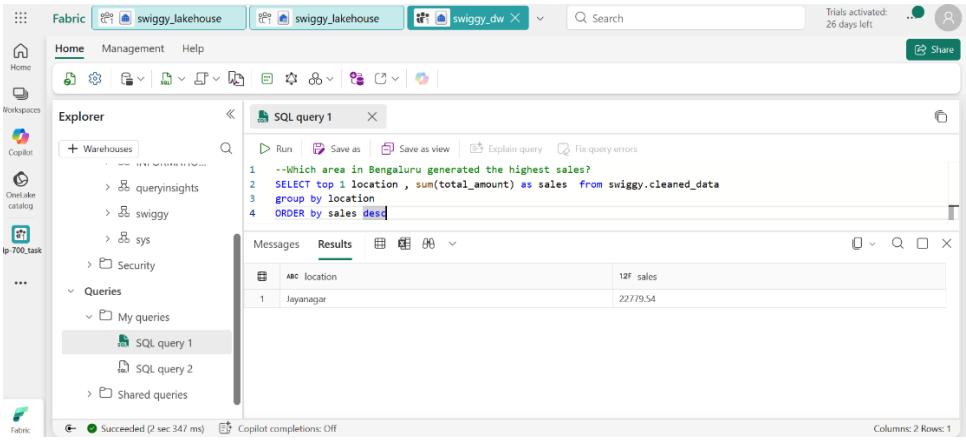
1. What is the **total revenue** generated across all orders?



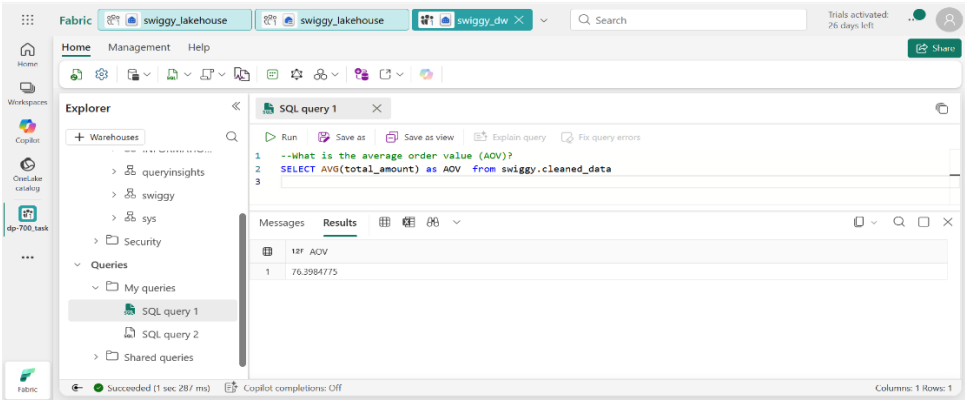
2. Which are the **top 5 cuisines by total revenue**?



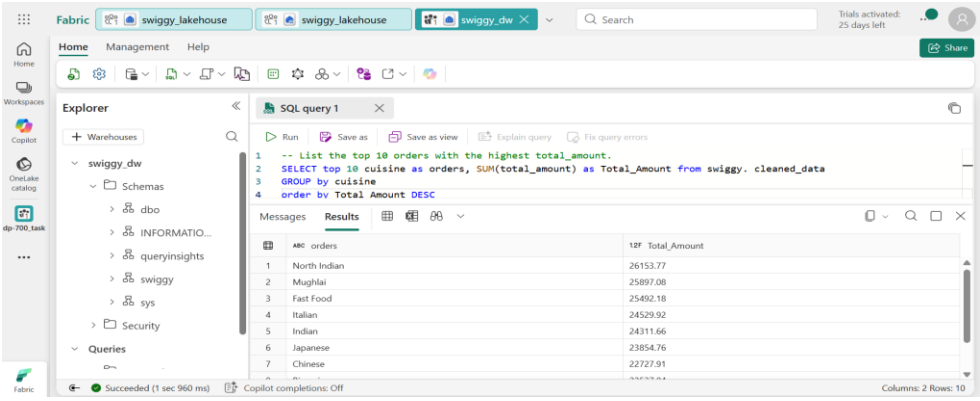
3. Which **area in Bengaluru generated the highest sales**?



4. What is the **average order value (AOV)**?

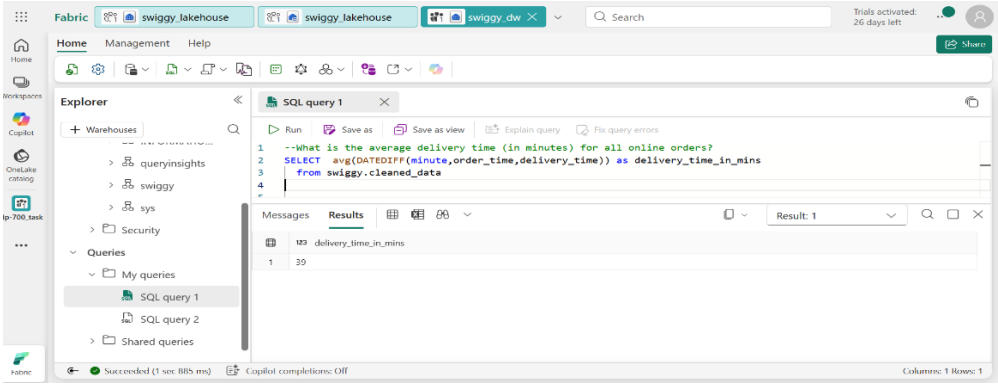


5. List the **top 10 orders with the highest total_amount**.

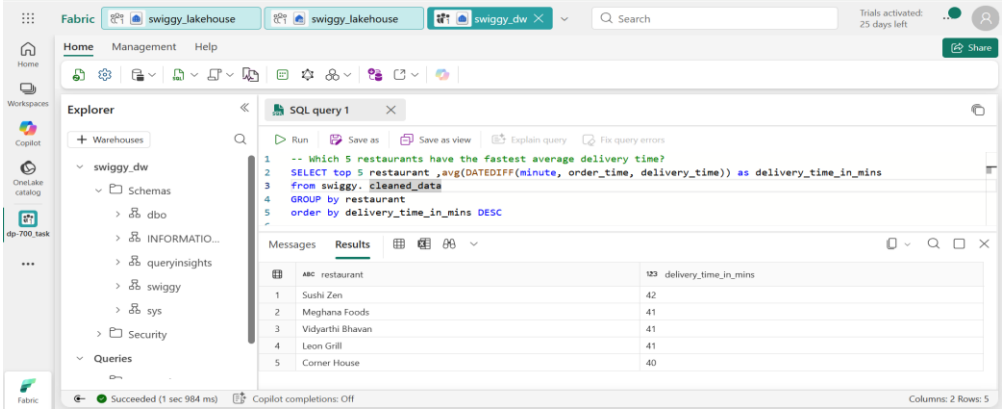


Category: Operational Efficiency

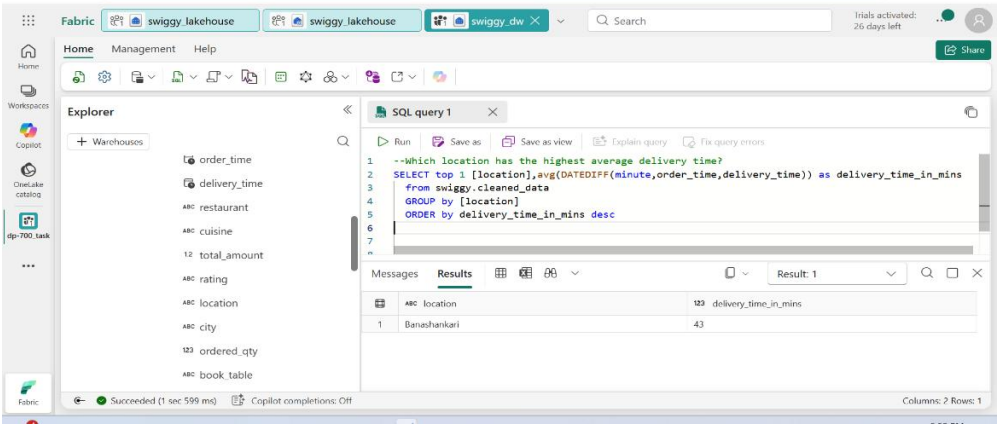
6. What is the **average delivery time (in minutes)** for all online orders?



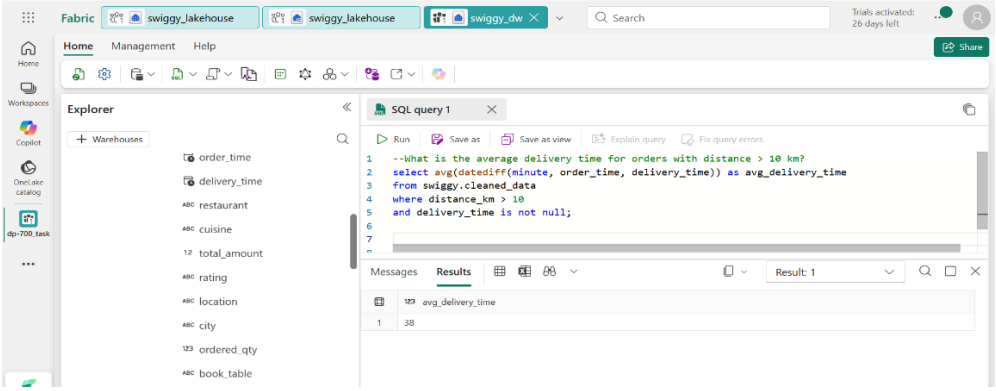
7. Which **5 restaurants have the fastest average delivery time**?



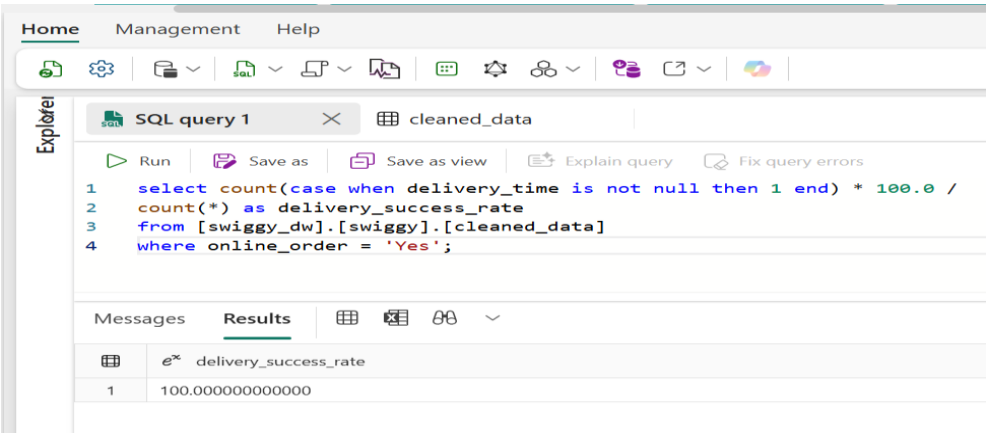
8. Which **location has the highest average delivery time**?



9. What is the **average delivery time for orders with distance > 10 km**?

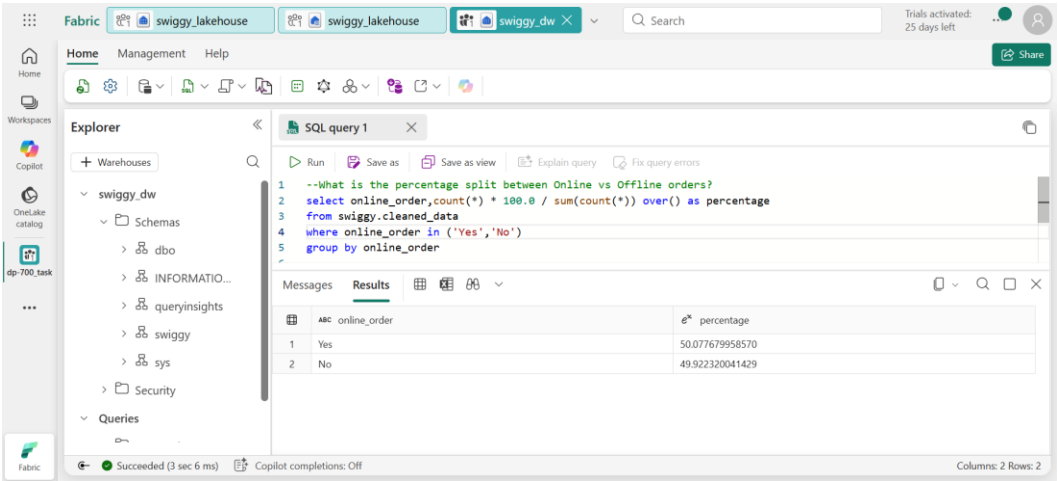


10. Delivery Success Rate: How many orders have a valid delivery_time compared to the total online_order = 'Yes' count?

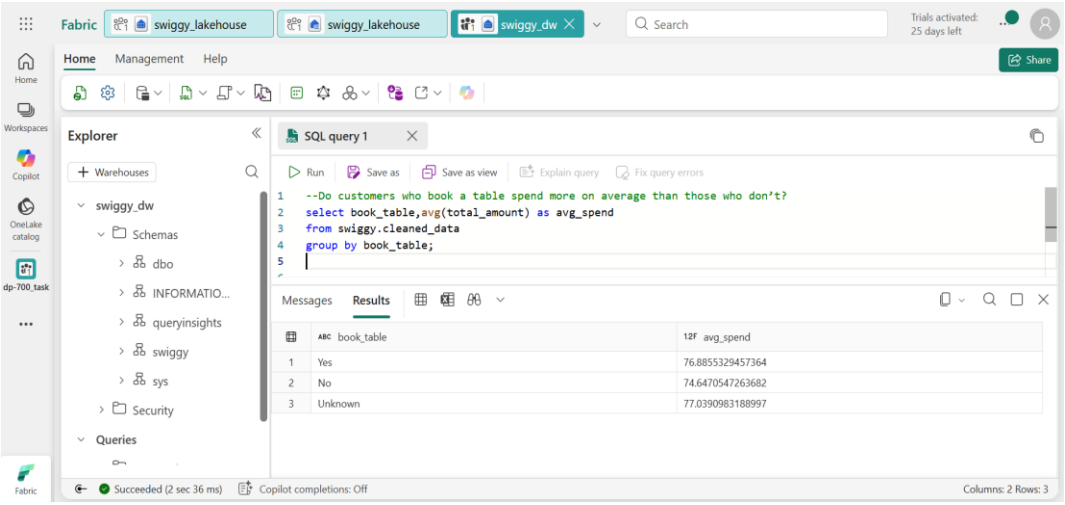


Category: Customer Behavior

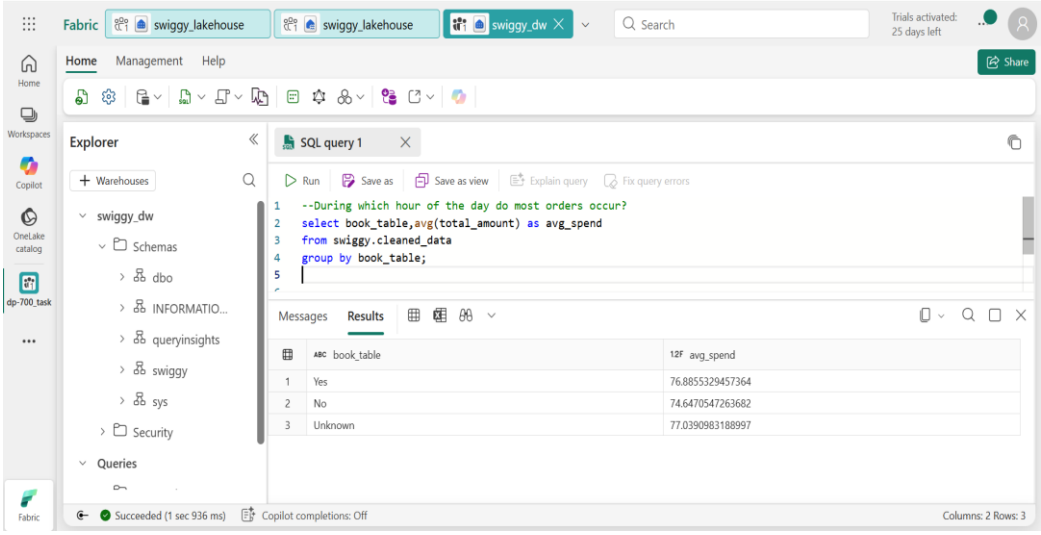
11. What is the **percentage split between Online vs Offline orders**?



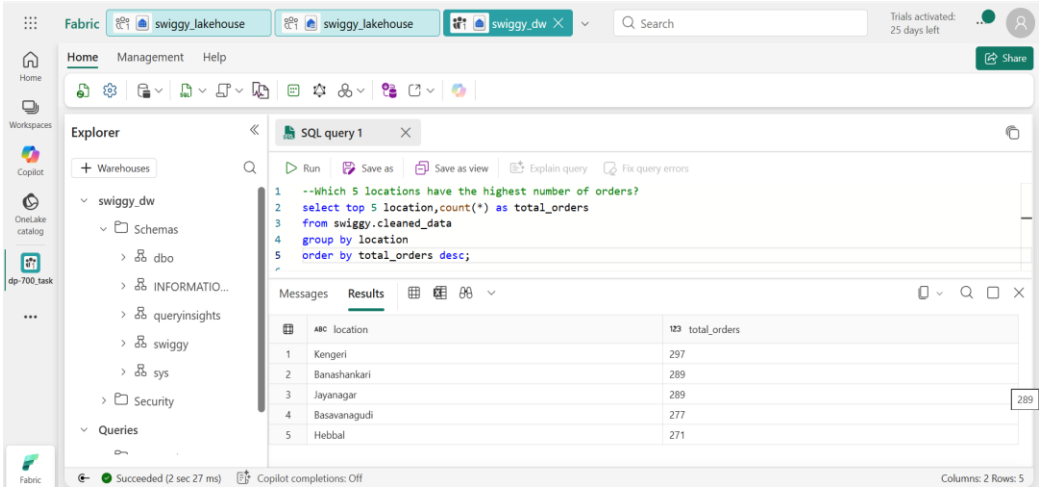
12. Do customers who **book a table spend more on average** than those who don't?



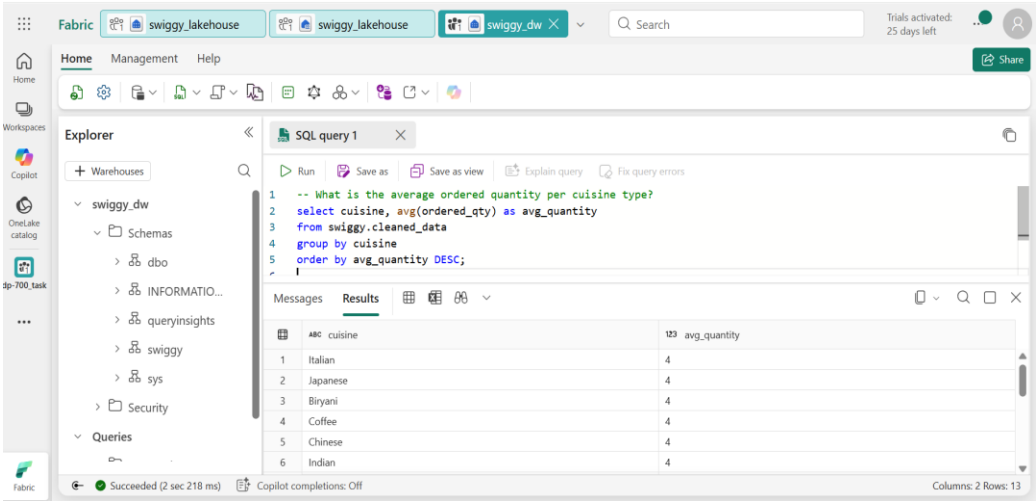
13. During which **hour of the day do most orders occur**?



14. Which **5 locations have the highest number of orders**?

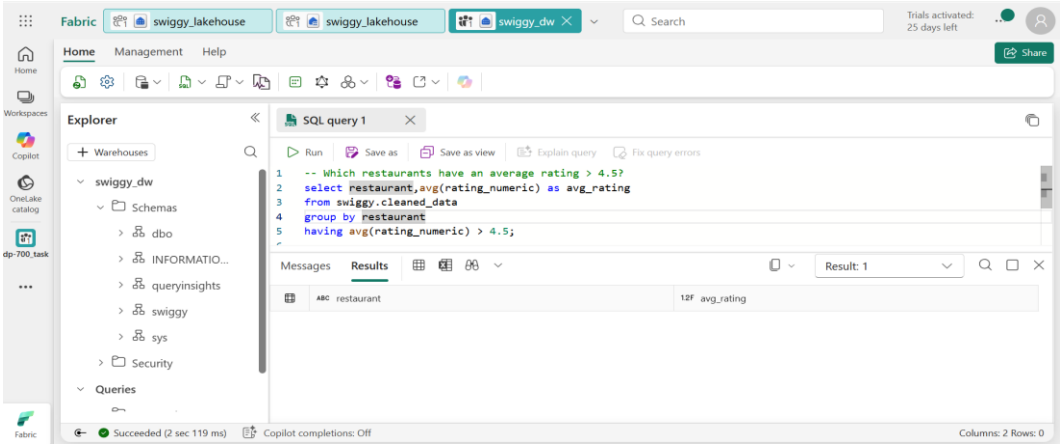


15. What is the **average ordered quantity per cuisine type**?

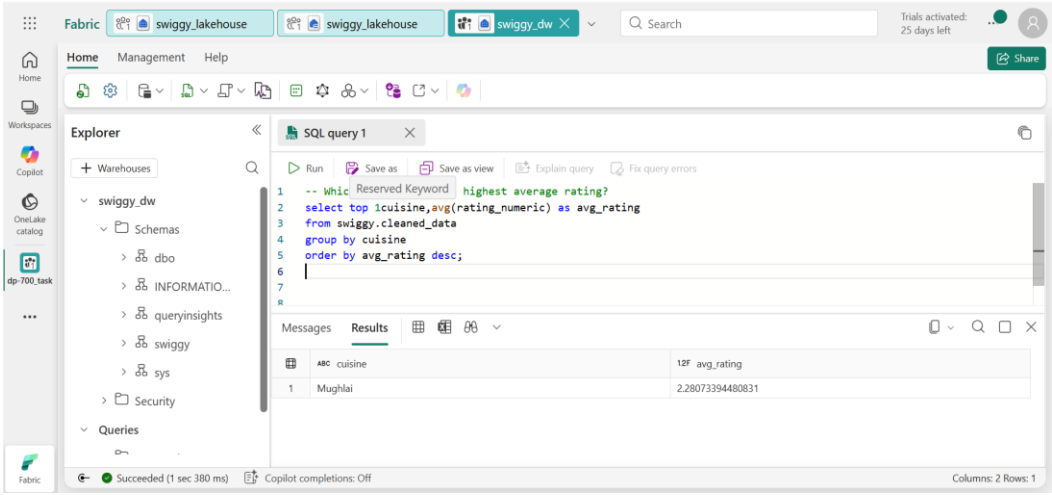


Category: Quality & Ratings

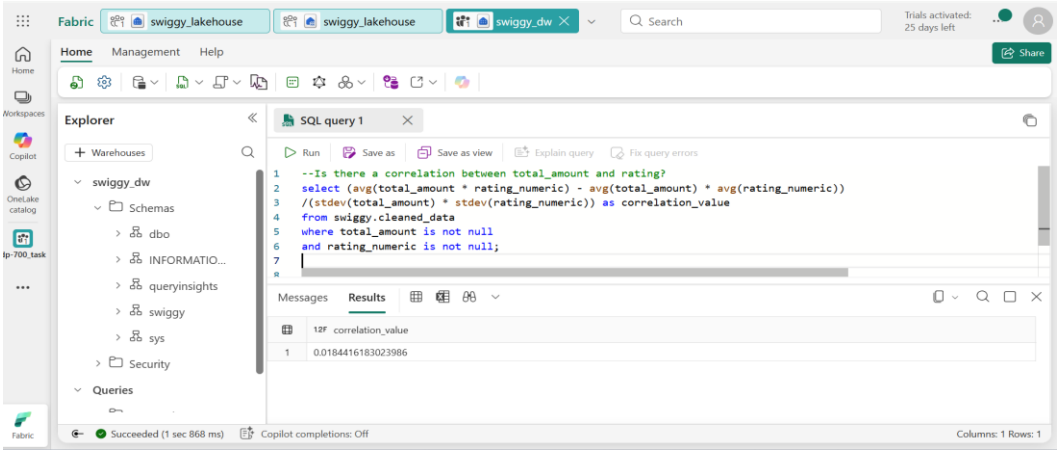
16. Which restaurants have an **average rating > 4.5**?



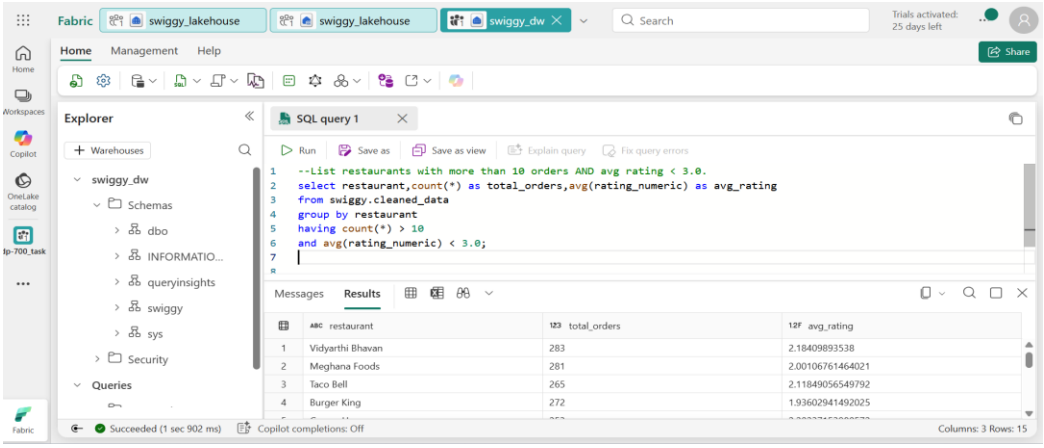
17. Which cuisine has the **highest average rating**?



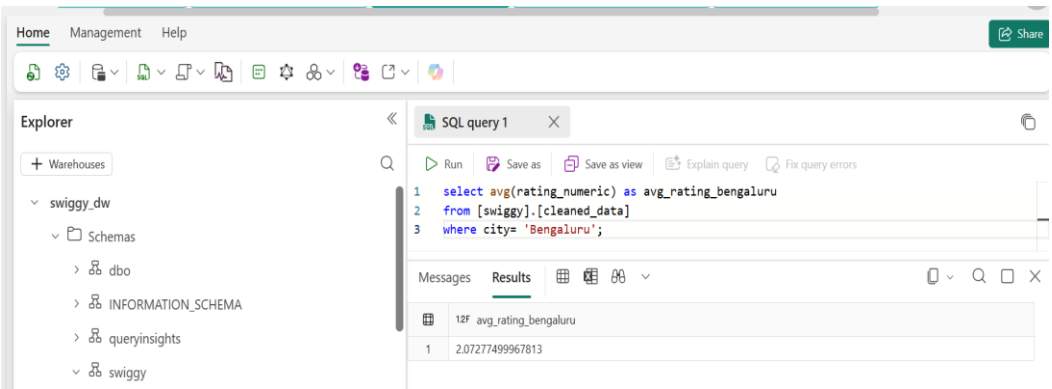
18. Is there a **correlation between total_amount and rating**?



19. List restaurants with **more than 10 orders AND avg rating < 3.0**.



20. What is the **average rating across Bengaluru**?



10. Conclusion

This project demonstrates a complete data engineering lifecycle including data ingestion, transformation, pipeline orchestration, data modeling, and analytical reporting using Microsoft Fabric.

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11. Tools & Technologies

- Microsoft Fabric
- PySpark
- SQL
- Delta Lake

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12. GitHub Repository

Sandeep2898/swiggy-fabric-project: Swiggy Data Engineering Project using Microsoft Fabric by ksr

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